

Autodesk® Scaleform®

Scaleform 最佳優化指南

Scaleform 4.0

Matthew Doyle

3.0

2011 4 25

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Autodesk® Scaleform® 4.2

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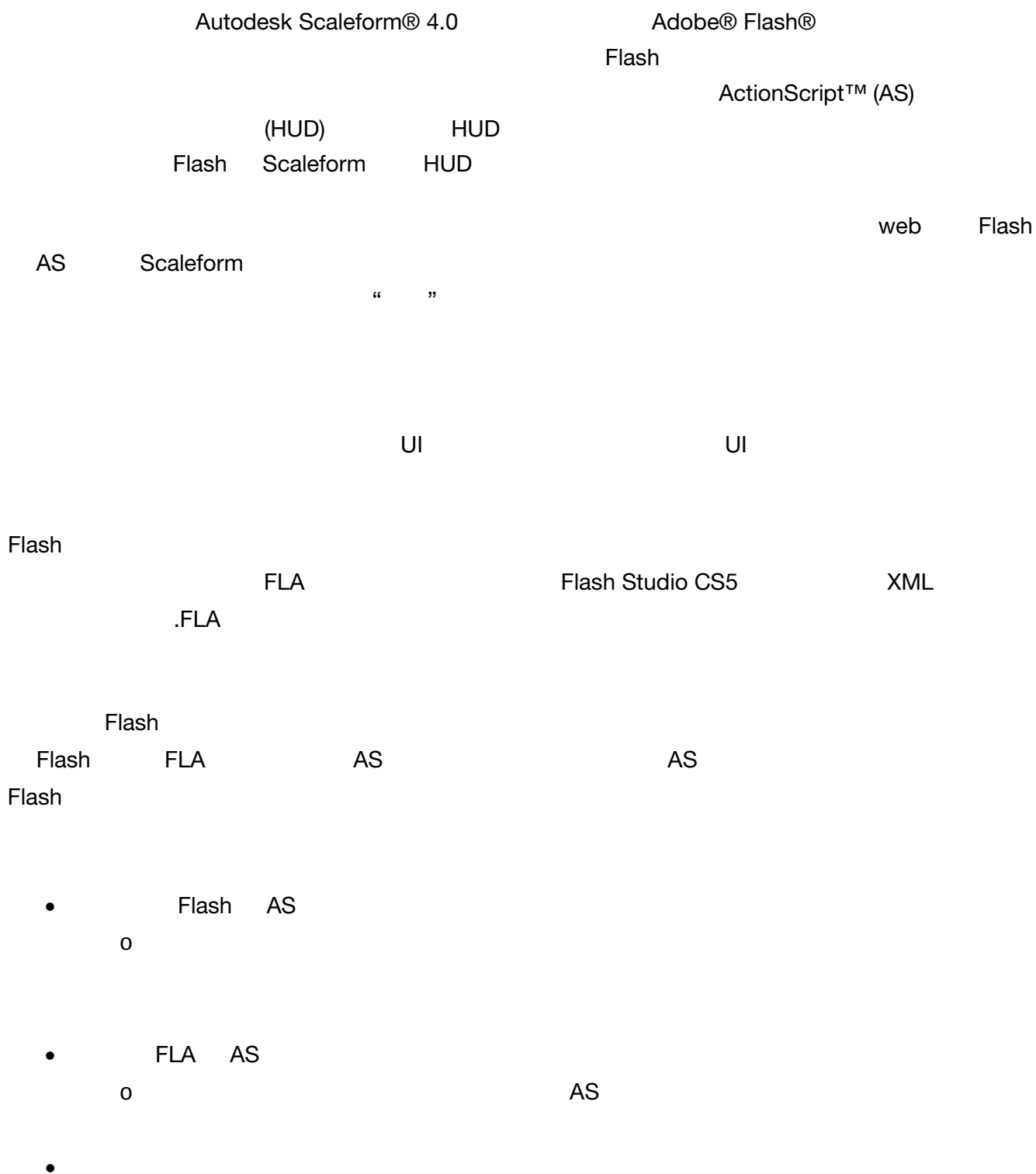
(301) 446-3200

(301) 446-3199

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1 概要



0

-

Flash AS

0

Flash

Flash

Flash AS

Flash

-

-

ActionScript

Flash

Flash

1.1

-

-

UI

Flash

Scaleform 3.3

Scaleform

(CLIK™)

1.2

-

-

Flash

HUD

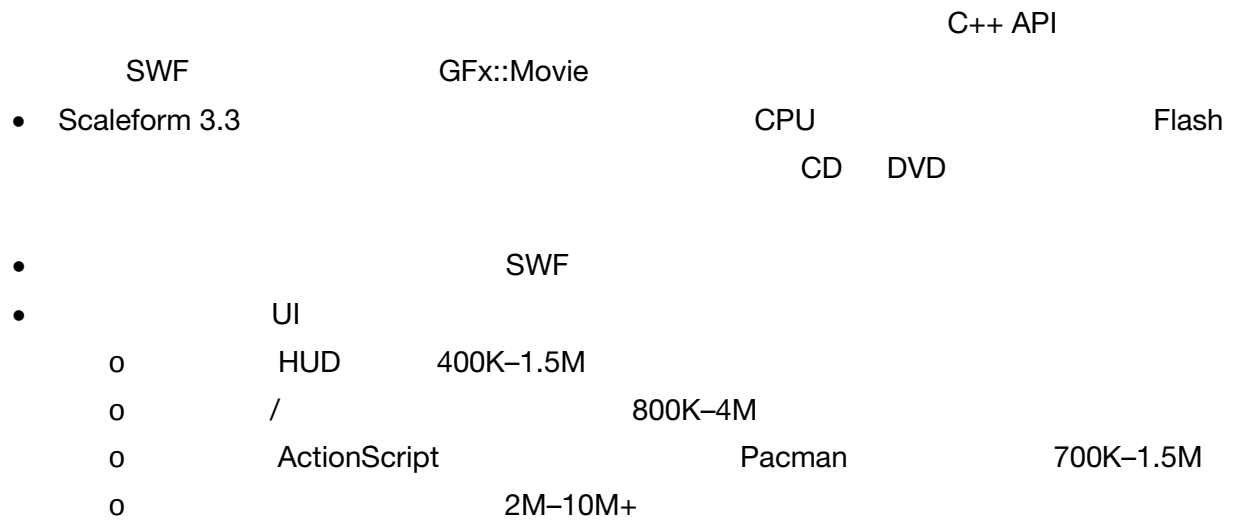
Flash

Flash

MMORPG

在 AS 端，不同文件可以通过

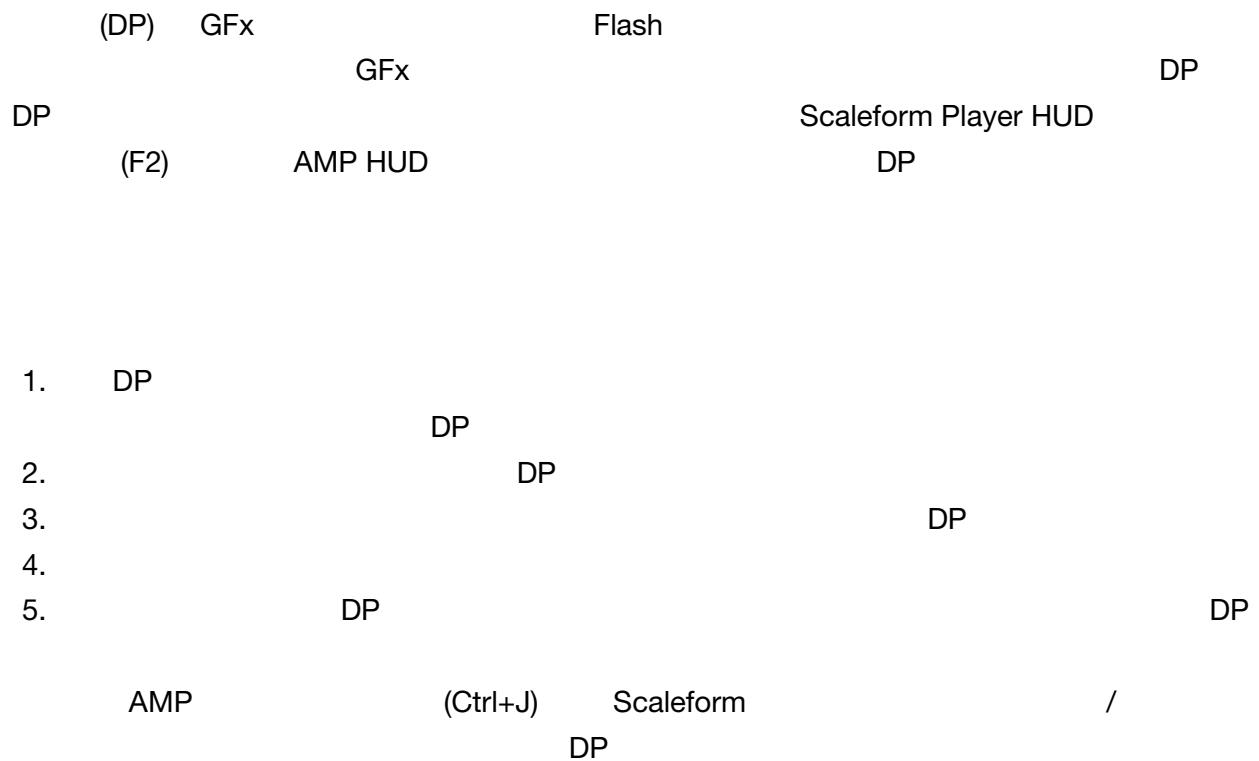
AS2 和 AS3 中的各种 SWF 加载 API 进行加载。



2

Flash

2.1



2.2

- 1.
- 2.
3. `_visible=false` `_alpha=0_alpha=0` (in AS2)
and `visible=false` rather than `alpha=0` (in AS3). `" stop()`

2.3

2.3.1

Flash

Scaleform

2.3.2

-
-

0

0

alpha Flash

`_visible`(AS2) or `visible` (AS3) `false`

`_alpha_alpha` (AS2) or `alpha` (AS3) 0 `_alpha alpha` property

100

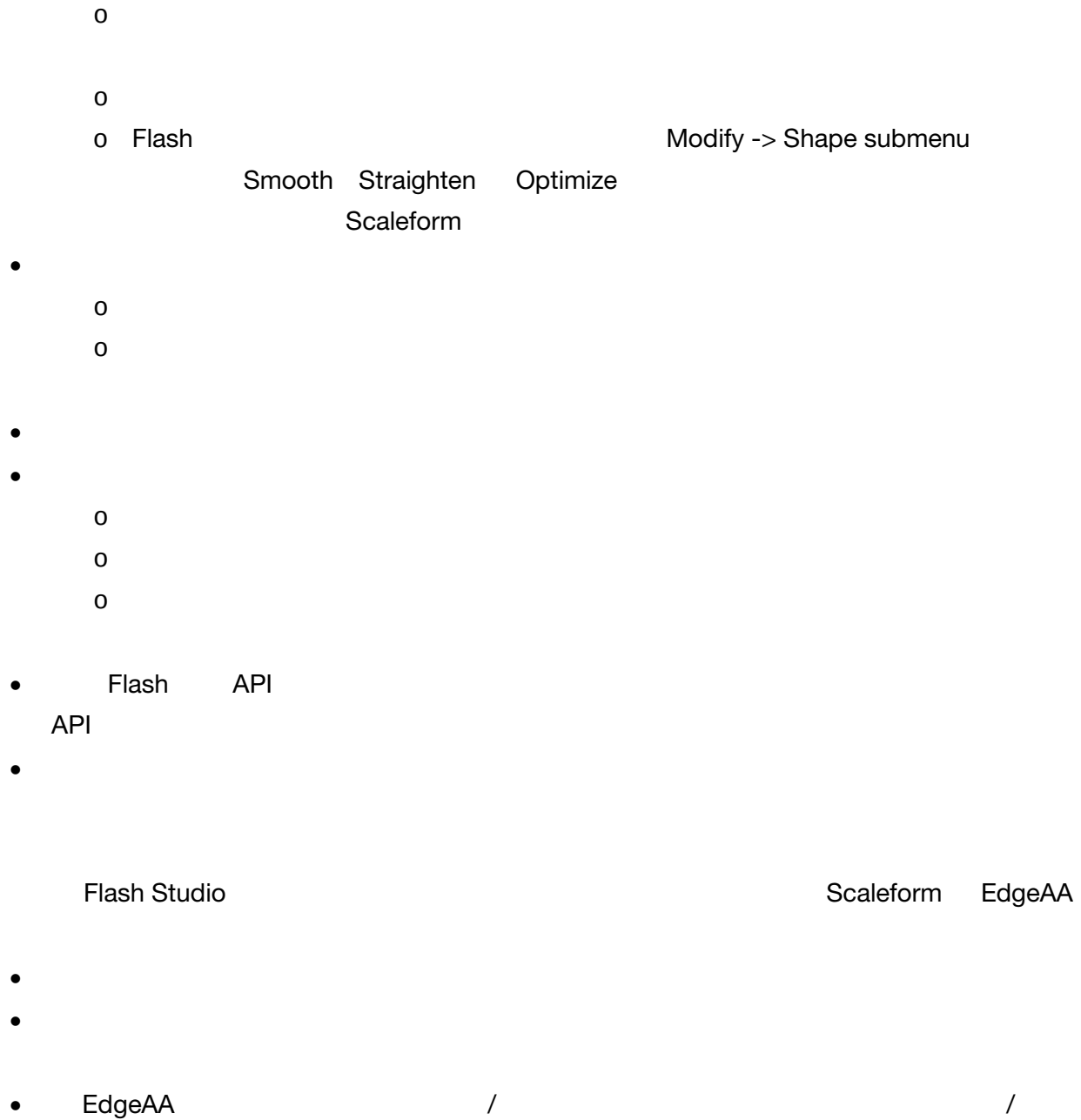
Flash

`_x` `_y`(in AS2)

or x and y properties (in AS3)

`_visible` / `visible` `false` Flash

-



2.3.3

Flash

SWF

`_visible`

false

_alpha

0

Scaleform

_alpha

0

false

CPU

ActionScript

CPU

SWF

`_visible`

false

- $\frac{1024 \times 1024}{512 \times 32} \times 2 = 16 \times 32 = 256 \times 128$

- JPEG

JPEG

JPEG

gfxexport

DXT

SWF

GFX

DXT

gfxexport

$-qp$ $-qh$

DDS

• /

- Edge AA

- ActionScript

- SWF/GFX

ActionScript

SWF

1024 x 1024

SWF

JPEG

4 MB

- UI 4

gfxexport

–i DDS

AMP

- CPU

- gradient.swf SWF gfxexport
-d0 gradients.swf SWF

-
-
-
-
-

Adobe Photoshop® Flash
 PNG
 Flash

2.3.4

FLA SWF
 AS
 12 24 FPS 12 FPS 12 FPS
 24 FPS 12 FPS
 5

[onEnterFrame](#) [onEnterFrame](#) (in AS2) or [Event.ENTER_FRAME](#) (in AS3)

[onEnterFrame/Event.ENTER_FRAME](#)

[setInterval](#)
[onEnterFrame/Event.ENTER_FRAME](#) [setInterval](#)

30 40 FPS CPU
 Flash UI

-
-
-

[true/false](#) [on/off](#) [_visible/visible](#) =
 /
 tween tween tween /
 0

- o tween Flash Tween
- o Scaleform CLIK Tween gfx.motion.Tween in AS2 or
scaleform.clik.motion.Tween in AS3 Flash Tween

- 60 Flash 60 FPS Flash

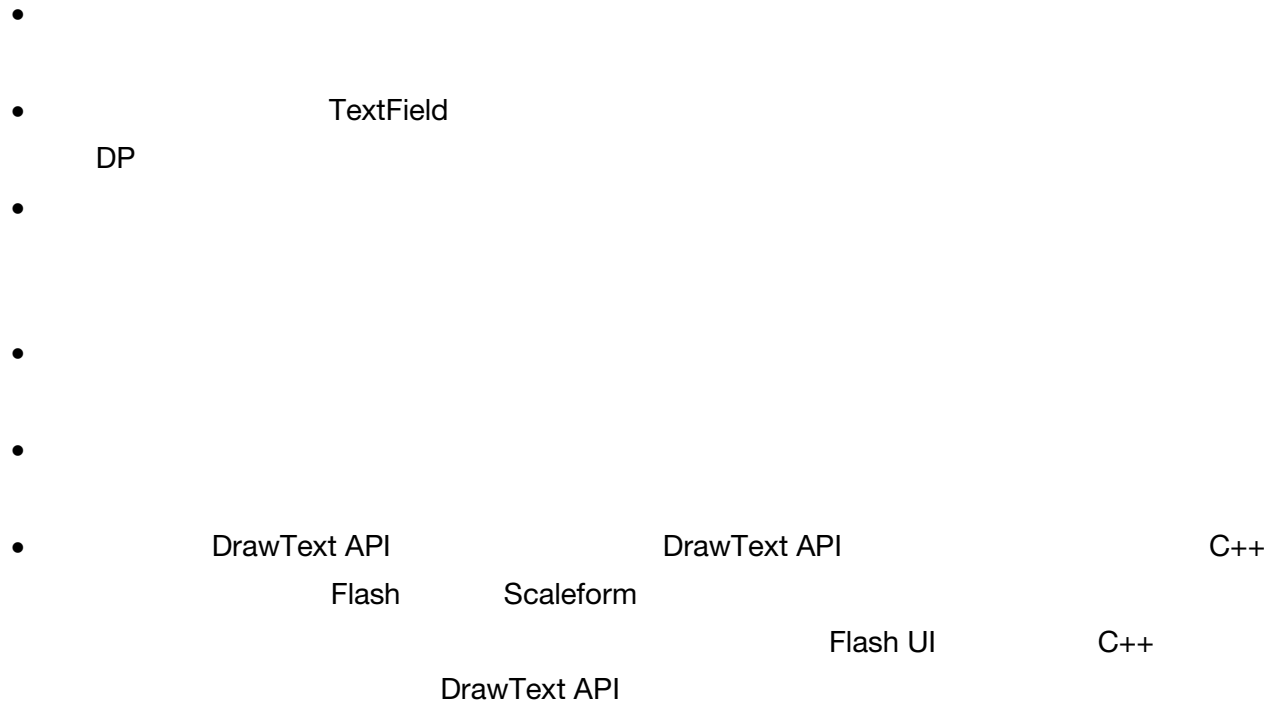
-
-
-
- /
- tween “ ”
- CPU
- morphing CPU

/

-

2.3.5

- SlotHeight gfxexport 48
DP
- DP
-
- 30 FPS
- ("TextField.variable")
- htmlText HTML
- gfxexport -fc, -fcl, -fcm



3 ActionScript

ActionScript

AS

—

—

AS

3.1 *ActionsScript2*

AS

- AS

AS

`attachMovie`

FLA

- AS

•

•

- if

- `on()` `onClipEvent()` `onEnterFrame` `onPress`

- AS Flash AS

`(onClipEvent, on)`
`(gotoAndPlay play stop )`

- 避免在包含多個動畫的長時間軸電影剪接中使用“遠距”`gotoAndPlay/gotoAndStop`。

o `gotoAndPlay/gotoAndStop`,
`,gotoAndPlay/gotoAndStop`,
`gotoAndPlay/gotoAndStop`

- 0 gotoAndPlay/gotoAndStop, ,
 , gotoAndPlay/gotoAndStop
- gotoAndPlay/gotoAndStop
 , gotoAndPlay/gotoAndStop ,
 ,
 gotoAndPlay/gotoAndStop
- C++
[Gfx::Movie::SetVariableArray](#) C++ AS
- Scaleform
Scaleform
- 0 bug
- eval()
- Array.length myArr.length

```
var fontArr:Array = TextField.getFontList();
var arrayLen:Number = fontArr.length;
for (var i:Number = 0; i < arrayLen; i++) {
    trace(fontArr[i]);
}
```

```
var fontArr:Array = TextField.getFontList();
for (var i:Number = 0; i < fontArr.length; i++) {
    trace(fontArr[i]);
}
```

- null
- removeListener()
- ,AS VM ,
 ,AS "if" ,
 , "com.xxx.yyy.aaa.bbb",VM "com" "xxx" "yyy" "aaa" "bbb",

```

        , "if" / /
        , ( "com", "xxx", "xxx"
"yyy", ) , Flash SWF
( c58923409876.functionName())

```

- SWF AS SWF
- AS
- SWF `trace()` Publish Settings
Flash Omit Trace Actions

-
- SWF AS `foo.bar.CustomClass`
SWF SWF

```

delete
delete foo.bar.CustomClass

```

- 100%
- `onEnterFrames`
- 0
Scaleform

3.1.1

-
-
-
-
- 0

3.1.2

-
- with

3.1.3 /

-
- var var ActionScript
-
-
- null SWF null
- AS

3.2 一般 ActionScript 3 指导原则

AS3

3.2.1 严格数据类型确定

-
- (Object) AS3 “
- (Array) (Vector)
- 4294967295 /
- Vector<*>
- Vector<*>
- Vector<int> Vector<String> Vector<YourFavoriteClass>
- Vector<*>
- flash.utils.Dictionary
- Gfx Flash

- / AS2 AS3
AS3

3.3

1. AS `onEnterFrame/ Event.ENTER_FRAME`
2. UI
3. `(_visible(AS2) or visible (AS3) true`
`stop() ) Advance`
4. `_global.noInvisibleAdvance(AS2) or`
`scaleform.gfx.Extensions.noInvisibleAdvance (AS3)`
`Advance true"`
`Advance`
- Flash Flash
`_global.gfxExtensions true Scaleform`
- 5.
- 6.

3.4 *onEnterFrame and Event.ENTER_FRAME*

- `onEnterFrame(AS2) or Event.ENTER_FRAME (AS3)`
`onEnterFrame/Event.ENTER_FRAME`
`setInterval setTimeout setInterval`
- `clearInterval`
- `setInterval setTimeout onEnterFrame/Event.ENTER_FRAME`
`onEnterFrame/Event.ENTER_FRAME`

3.4.1 清理 onEnterFrame

delete onEnterFrame

```
delete this.onEnterFrame;  
delete mc.onEnterFrame;
```

null undefined onEnterFrame (this.onEnterFrame = null;)
onEnterFrame Scaleform
onEnterFrame

3.5 ActionScript 2 Optimizations

3.5.1 onClipEvent 和 on Events

onClipEvent() on() onEnterFrame onPress

-
- onClipEvent
this _global arguments super 256

onLoad
onClipEvent(construct) onEnterFrame

```
onClipEvent(construct)  
{  
    this.onLoad = function()  
    {  
        //  
    }  
}
```

onClipEvent(load)

3.5.2 Var 關鍵字

var

AS

var

AS

```

var i = 1000;
countIt = function()
{
    num = 0;
    for(j=0; j<i; j++)
    {
        j++;
        num += Math.random();
    }
    displayNumber.text = num;
}

```

```

var i = 1000;
countIt = function()
{
    var num = 0;
    var ii = i;
    for(var j=0; j<ii; j++)
    {
        j++;
        num += Math.random();
    }
    displayNumber.text = num;
}

```

3.5.3 預存

var

```

function foo(var obj:Object)
{
    for (var i = 0; i < obj.size; i++)
    {
        obj.value[i] = obj.num1 * obj.num2;
    }
}

```

```
function foo(var obj:Object)
{
    var sz = obj.size;
    var n1 = obj.num1;
    var n2 = obj.num1;
    for (var i = 0; i < sz; i++)
    {
        obj.value[i] = n1*n2;
    }
}
```

```
var floor = Math.floor
var ceil = Math.ceil
num = floor(x) - ceil(y);
```

```
var keyDown = Key.isDown;
var keyLeft = Key.LEFT;
if (keyDown(keyLeft))
{
    //
}
```

3.5.4 預存長路徑

```
mc.ch1.hc3.djf3.jd9._x = 233;
mc.ch1.hc3.djf3._x = 455;
```

```
var djf3 = mc.ch1.hc3.djf3;
djf3._x = 455;

var jd9 = djf3.jd9;
jd9._x = 223;
```

3.5.5 複雜運算式

C

```
this[_global.mynames[queue]][_global.slots[i]].gosplash.text =
_global.MyStrings[queue];
```

```
var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
var slot_i = _global.slots[i];
_splqueue[slot_i].gosplash.text = _splstring;
```

```
for(i=0; i<3; i++)
{
    this[_global.mynames[queue]][_global.slots[i]].gosplash.text =
        _global.MyStrings[queue];
    this[_global.mynames[queue]][_global.slots[i]].gosplash2.text =
        _global.MyStrings[queue];
    this[_global.mynames[queue]][_global.slots[i]].gosplash2.textColor =
        0x000000;
}
```

```
var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
for (var i=0; i<3; i++)
{
    var slot_i = _global.slots[i];
    _splqueue[slot_i].gosplash.text = _splstring;
    _splqueue[slot_i].gosplash2.text = splstring;
    _splqueue[slot_i].gosplash2.textColor = 0x000000;
}
```

```
var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
for (var i=0; i<3; i++)
{
    var slot_i = _global.slots[i];
    var elem = _splqueue[slot_i];
    elem.gosplash.text = _splstring;
    var gspl2 = elem.gosplash2;
    gspl2.text = splstring;
```



```
    gsp12.textColor = 0x000000;  
}
```

4 HUD

Scaleform

(HUD)

Scaleform

HUD

HUD

4.1

HUD

HUD

HUD

C++

4.1 SWF

HUD

80K

HUD

SWF

1.

Advance

HUD

Flash

AS

Flash

Flash

Advance

HUD

HUD

Advance

Scaleform

Gfx::Movie

Advance

Advance

2.

HUD

HUD

4.2

SWF

Flash AS `loadMovie`

Flash

1. `_visible = false`(AS2) or `visible = false` (AS3)

`_global.noInvisibleAdvance`(AS2) or

`scaleform.gfx.Extensions.noInvisibleAdvance` (AS3) `true`

HUD

`_visible/visible = false`

Advance

Flash

HUD

/ HUD

Advance

`_global.gfxExtensions`(AS2) or `scaleform.gfx.Extensions.enabled`

(AS3) `true` Scaleform

2. `SetVariableArray`

`SetVariableArray`

3. HUD `onEnterFrame` (AS2) or `Event.ENTER_FRAME` (AS3)

HUD `onEnterFrame//Event.ENTER_FRAME`

Advance

4. AS

Flash

AS

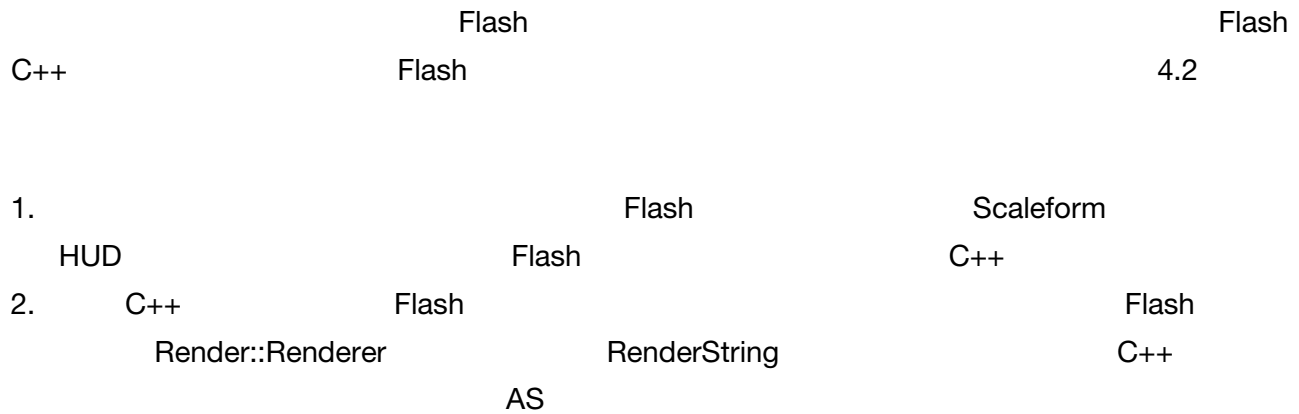
30-40 FPS

15-20 FPS

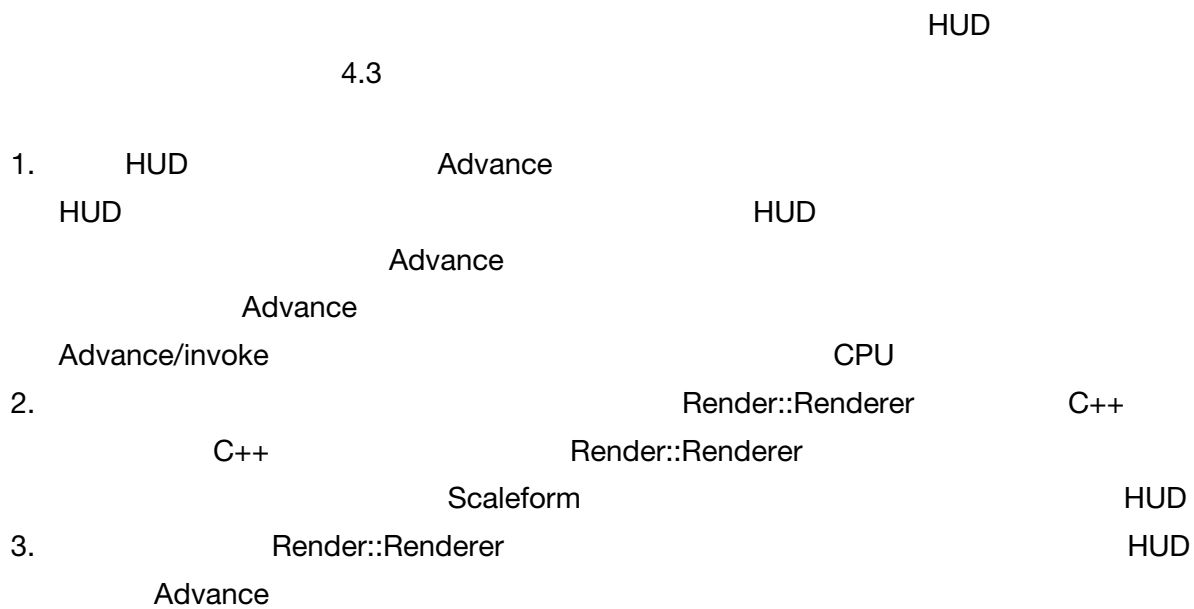
HUD

5.

4.3



4.4 (Advanced)



4.5 *Flash* *HUD*

HUD C++ Scaleform Flash
 HUD Flash HUD Scaleform Player
 Scaleform
 C++ HUD Scaleform
 /
 DP Scaleform
 HUD Scaleform
 Scaleform /
 DrawText API C++ Scaleform
 Flash Scaleform HUD
 /
 " FAQ " / "DrawText API "
 HUD
 •
 •
 • PC Wii™
 o Gfx::Movie::SetMouseCursorCount(0);
 o
 • / HUD noInvisibleAdvance
 noInvisibleAdvance (AS2) or
 scaleform.gfx.Extensions.noInvisibleAdvance (AS3
 _visible(AS2) or visible (AS3) = false

-

-

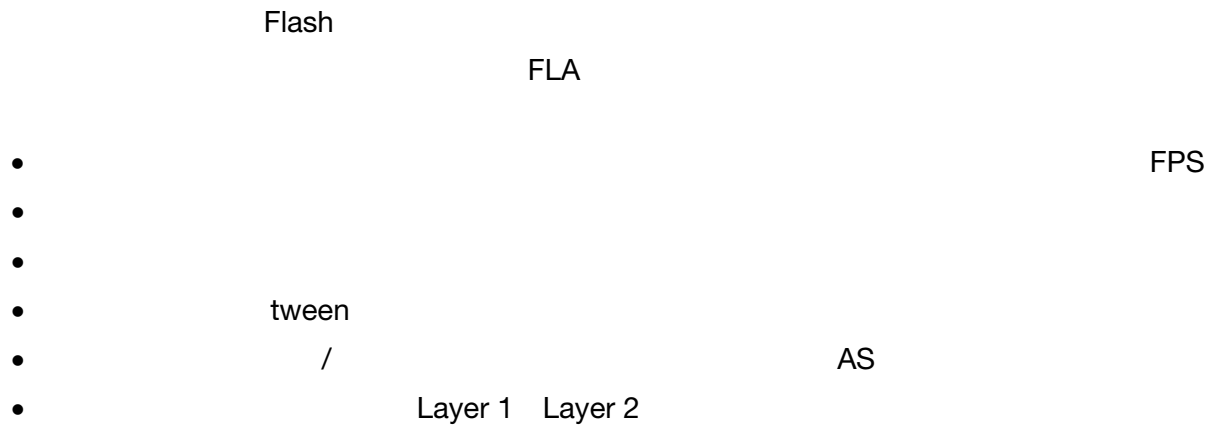
[2.1](#)

[" "](#)

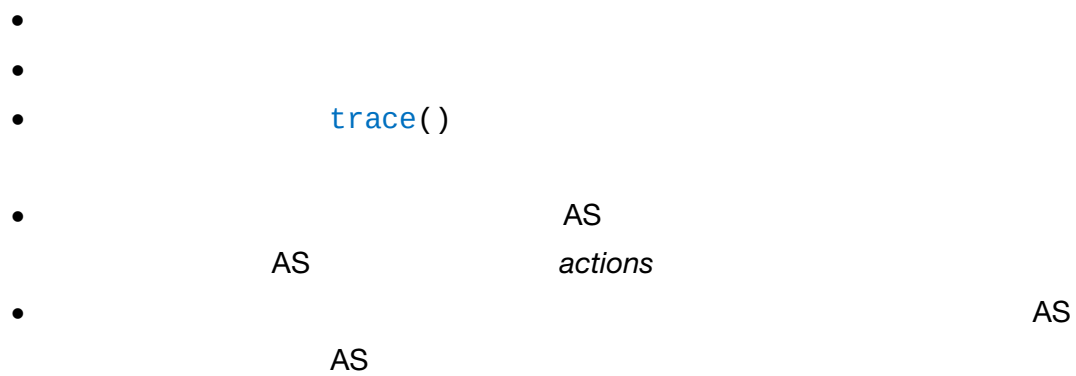
FAQ

5

5.1 *Flash*



5.2



6

ActionScript 2.0

http://www.adobe.com/devnet/flash/articles/as_bestpractices.html

Flash 8

http://www.adobe.com/devnet/flash/articles/flash8_bestpractices.html

Flash ActionScript 2.0

http://www.adobe.com/devnet/flash/articles/actionscript_guide.html